

B.Sc. Semester IV Examination 2025-2026
STATISTICS
STAMN-42 : Practical based on STAMN-41

Time: 3 Hours

Max. Marks: 70

Instructions: Attempt any 3 questions. Each question carries equal marks.

Q.1 A quality control engineer at a textile factory measures the breaking strength (in Newtons) of fabric samples. A random sample of $n = 36$ observations is drawn from a normal population with mean $\mu = 42.4$ N and known variance $\sigma^2 = 16$ N.

- (a) Construct confidence intervals for the population mean at 70%, 80%, 90%, and 95% levels.
- (b) Interpret each confidence interval in context.
- (c) Compute the length of each confidence interval.
- (d) Plot the relationship between confidence level and interval length.
- (e) The factory's safety standard requires breaking strength to be at least 41 N. Based on your 95% confidence interval, does the fabric meet the standard? Interpret your result in context.

Q.2 A pharmaceutical company wants to investigate whether a new dietary supplement reduces fasting blood sugar levels (in mg/dL) in diabetic patients. The blood sugar levels of 20 patients were recorded before and after 8 weeks of consuming the supplement:

oprule Patient	1	2	3	4	5	6	7	8	9	10
Before	145	162	138	175	153	168	140	159	172	148
After	138	150	135	160	145	155	136	148	158	140
ottomrule										
oprule Patient	11	12	13	14	15	16	17	18	19	20
Before	165	155	142	178	160	135	150	170	143	158
After	152	143	138	162	149	130	142	155	138	147
ottomrule										

Test at 5% level of significance whether the dietary supplement significantly reduced the blood sugar levels of patients.

Q.3 A market research firm surveyed 10,000 customers from two different cities — Mumbai and Delhi — and recorded their preferred shopping time (Hour 0 to Hour 9 representing 10 different time slots of the day). The frequency of customers preferring each time slot is as follows:

oprule Time Slot	0	1	2	3	4	5	6	7	8	9
Mumbai	983	1024	1015	978	1032	1008	965	1021	994	980
Delhi	1002	988	975	1034	998	1016	1025	979	1003	980
ottomrule										

Using the above data, examine whether the distribution of preferred shopping time slots is independent of the city (Mumbai or Delhi) to which the customers belong. Formulate appropriate hypotheses and apply an appropriate test at the 5% level of significance. Based on your analysis, state whether there is any significant difference in the shopping time preferences of customers from Mumbai and Delhi. Interpret your result in context.

Q.4 A meteorological department recorded the daily maximum temperature (in °C) in a city over a period of 20 day during the monsoon season to check whether the temperature fluctuations occur in a random order. The recorded temperatures are as follows:

28 31 27 33 29 30 26 32 28 31
27 33 30 28 31 26 29 32 27 30

Test at 5% level of significance that the daily maximum temperatures occur in a random order.